```
clear;
clc;
close all;

% Design Parameters
Fs = 8000;
Ap = 1;
As = 40;

% 1. Low-Pass Filter Design
fp_lp = 1000;
fs_lp = 1500;

wp_lp = fp_lp / (Fs/2);
ws_lp = fs_lp / (Fs/2);

[N_lp, Wn_lp] = buttord(wp_lp, ws_lp, Ap, As);
[b_lp, a_lp] = butter(N_lp, Wn_lp, 'low');

% 2. High-Pass Filter Design
fp_hp = 2500;
fs_hp = 1800;

wp_hp = fp_hp / (Fs/2);
ws_hp = fs_hp / (Fs/2);

[N_hp, Wn_hp] = buttord(wp_hp, ws_hp, Ap, As);
[b_hp, a_hp] = butter(N_hp, Wn_hp, 'high');

% Display Numerical Outputs
fprintf('=== BUTTERWORTH IIR FILTERS ===\n\n');
```

```
=== BUTTERWORTH IIR FILTERS ===
```

```
fprintf('--- LOW-PASS FILTER ---\n');
```

```
--- LOW-PASS FILTER ---
```

```
fprintf('Filter Order: %d\n', N_lp);
```

```
Filter Order: 12
```

```
fprintf('Cutoff Frequency: %.2f Hz\n', Wn_lp*(Fs/2));
```

```
Cutoff Frequency: 1087.84 Hz
```

```
fprintf('Numerator Coefficients:\n');
```

```
Numerator Coefficients:
```

```
disp(b_lp);
```

```
    0.0000    0.0000    0.0002    0.0006    0.0013    0.0021    0.0024    0.0021    0.0013    0.0006    0.
```

```
fprintf('Denominator Coefficients:\n');
```

```
Denominator Coefficients:
```

```
disp(a_lp);
```

```
    1.0000   -5.4622   14.6771  -25.1616   30.3497  -26.9537   17.9862   -9.0534    3.4018   -0.9283    0.
```

```
fprintf('--- HIGH-PASS FILTER ---\n');
```

```
--- HIGH-PASS FILTER ---
```

```
fprintf('Filter Order: %d\n', N_hp);
```

```
Filter Order: 10
```

```
fprintf('Cutoff Frequency: %.2f Hz\n', Wn_hp*(Fs/2));
```

```
Cutoff Frequency: 2379.76 Hz
```

```
fprintf('Numerator Coefficients:\n');
```

```
Numerator Coefficients:
```

```
disp(b_hp);
```

```
    0.0006   -0.0055    0.0248   -0.0662    0.1159   -0.1390    0.1159   -0.0662    0.0248   -0.0055    0.
```

```
fprintf('Denominator Coefficients:\n');
```

```
Denominator Coefficients:
```

```
disp(a_hp);
```

```
    1.0000    1.8915    2.8535    2.6014    1.8704    0.9534    0.3726    0.1030    0.0199    0.0023    0.
```

```
% Frequency Response
[H_lp, f] = freqz(b_lp, a_lp, 1024, Fs);
[H_hp, ~] = freqz(b_hp, a_hp, 1024, Fs);

% Avoid log of zero
mag_lp = 20*log10(max(abs(H_lp), 1e-10));
mag_hp = 20*log10(max(abs(H_hp), 1e-10));

% Plot Responses
figure('Name', 'Butterworth IIR Filters');

subplot(2,2,1);
plot(f, mag_lp, 'b', 'LineWidth', 1.5);
grid on;
title(['LPF Magnitude Response (N = ', num2str(N_lp), ')']);
xlabel('Frequency (Hz)');
```

```
ylabel('Magnitude (dB)');
ylim([-80 5]);

subplot(2,2,2);
plot(f, angle(H_lp)*180/pi, 'r', 'LineWidth', 1.5);
grid on;
title('LPF Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');

subplot(2,2,3);
plot(f, mag_hp, 'b', 'LineWidth', 1.5);
grid on;
title(['HPF Magnitude Response (N = ', num2str(N_hp), ')']);
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
ylim([-80 5]);

subplot(2,2,4);
plot(f, angle(H_hp)*180/pi, 'r', 'LineWidth', 1.5);
grid on;
title('HPF Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
```

